

Veer Patel

AI Research Engineer

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Summary

AI Research Engineer with 2+ years of experience building production-scale AI systems across AI drug discovery, computer vision, and large language models. Expertise in generative AI, transformer architectures, distributed multi-GPU training, and end-to-end model deployment, with a proven track record of developing scalable solutions for molecular design, medical imaging, and multi-agent AI systems.

Experience

Software Development Engineer (AI)

Jun 2024 – Present

Nuvo AI Pvt. Ltd., Vapi

- **Architected and delivered end-to-end AI platforms** across **AI drug discovery** and **medical imaging**, owning the complete lifecycle from problem definition and data strategy to model development, deployment, and production integration.
- Collaborated closely with **structural biologists** and **medicinal chemists** to translate domain knowledge into AI-driven molecular design workflows, developing generative pipelines for **pharmacophore-based molecular generation**, **fragment optimization**, **molecular docking**, and **lead prioritization**.
- Partnered with **orthopedic surgeons** and **data acquisition teams** to design clinically relevant AI solutions for trauma care, refining annotation strategies, defining diagnostic workflows, and ensuring high-quality medical imaging datasets for model development.
- Designed **modular, multi-stage AI pipelines** integrating data ingestion, preprocessing, distributed training, inference orchestration, and API services, enabling scalable and maintainable production AI systems.
- Led the technical architecture and end-to-end development of a **AI platform for orthopedic trauma** processing **1L+ X-ray images**, integrating object detection, transformer-based classification, and hierarchical inference for fracture detection and implant size recommendation.
- Engineered scalable **generative AI solutions** for structure-aware molecular design using **pharmacophore-conditioned molecular generation** and **fragment optimization** techniques to accelerate structure-based drug discovery workflows.
- Led a **cross-functional engineering team of 6 (3 Engineers + 3 Interns)**, driving technical planning, architecture decisions, code reviews, sprint execution, and successful delivery of production AI products.
- Built **distributed multi-GPU training infrastructure** using **PyTorch**, **DeepSpeed**, and **HuggingFace Accelerate**, significantly improving experimentation throughput and large-scale model training efficiency.
- Developed **production-ready backend services** using **FastAPI**, **Pydantic**, and **REST APIs**, enabling reliable deployment, model serving, and seamless integration with downstream applications.
- Designed **high-throughput data pipelines** leveraging **MinIO/S3**, intelligent caching, and fault-tolerant data loading to support scalable dataset management and reliable AI training workflows.
- Developed **multi-agent AI systems** leveraging **LLMs** and **VLMs** to automate fracture assessment, clinical recommendations, structured reporting, and intelligent decision-support workflows.

Product Owner & Data Scientist

Oct 2023 – Apr 2024

Tech Curve AI and Innovations, Bangkok, Thailand (Remote)

- Owned end-to-end **product strategy** for an **AI-driven HR automation platform**, defining UI/UX workflows, backend architecture, and feature roadmap
- Built an **LLM-powered resume ranking engine (LLaMA-based)**, reducing recruiter screening time by **~70%**
- Developed an **LLM-driven vendor recommendation system** integrated with **VMS platforms** across 10+ product categories
- Engineered **end-to-end NLP pipelines** for document parsing, semantic matching, and ranking, improving candidate-job alignment accuracy

Education

M.Tech in Computer Science Engineering (Specialization in AI & ML)

2021 – 2023

Ajeenkya D.Y. Patil University (CGPA: 9.02)

B.Tech in Electronics & Telecommunication Engineering

2017 – 2021

A.D. Patel Institute of Technology (CGPA: 7.43)

Projects

Stock Market Prediction (Master's Thesis) — Developed LSTM, Random Forest, and MLP models for financial time-series forecasting, achieving 75% accuracy on real-world datasets; implemented feature engineering and model evaluation pipelines for performance optimization

Equity Research AI Tool — Built a LangChain + Mistral-based system for extracting insights and enabling question-answering over financial news and reports; designed document ingestion and semantic retrieval pipeline for structured financial analysis

Skills

Languages: Python, SQL, JavaScript

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, Transformers, Distributed Training

Computer Vision: OpenCV, Object Detection, Image Classification, Segmentation

Generative AI & LLMs: LLaMA, HuggingFace, LangChain, RAG, PEFT, QLoRA

Systems & Backend: FastAPI, REST APIs, Model Serving, ONNX, NVIDIA Triton Inference Server

Tools & MLOps: Docker, Git, MLflow, Model Evaluation, BentoML, PyMOL

Cloud: AWS (S3, EC2), Azure AI, Google Cloud Platform (GCP)